Children Storytelling & Narration Recording Platform

1. Introduction

This document outlines the design and development of a web-based storytelling platform aimed at children. The platform encourages children to narrate stories in their own words by observing visual prompts (images) while recording their voice. The system is designed for educational and research purposes, enabling teachers and researchers to analyze narration skills, creativity, and cognitive patterns.

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2. Objectives

- Encourage creative storytelling among children.
- Collect narration data for research and educational analysis.
- Provide an intuitive, child-friendly user interface.
- Enable secure storage and management of recorded data.

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3. Target Users

Primary Users

- Children (ages 5–12)

Secondary Users

- Teachers
- Researchers
- Administrators (platform owners)

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4. System Overview

The platform consists of two main modules: 1. Child Interaction Module (Frontend Experience) 2. Admin Management Module (Backend Dashboard)

The backend will use MongoDB for storing metadata and references to recordings.

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5. User Flow

5.1 Child Flow

1. Open website

2. Enter name
3. Select a story
4. View story images and narrate
5. Record voice
6. Submit or cancel recording
7. Recording saved locally and on server

5.2 Admin Flow

1. Click Admin Login
2. Authenticate
3. View all recordings
4. Play / Download / Delete recordings
5. Add new stories

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6. Features

6.1 Home Page

- Clean, colorful, child-friendly UI
- Input field to enter child's name
- "Start" button to proceed
- "Admin Login" option at top corner

6.2 Story Selection Page

- Displays 2–3 (or more) stories
- Each story represented with:
- Title
- Cover image
- Click to select story

6.3 Story Narration Page

Layout

Center Panel - Displays one image at a time (out of 6) - Navigation buttons: - Next - Previous

Side Panel - Thumbnails of all 6 images - Click to jump to any image

Top Corner Options - Instructions (how to narrate) - Characters (list of characters) - Summary (brief overview of story)

6.4 Recording Module

Located at the bottom of the screen: - Start Recording - Pause Recording - Stop Recording

Recording Flow

1. User clicks Start
2. Narrates story while viewing images
3. Can pause/resume
4. Stops recording at the end

Post Recording Options

- Submit Recording
- Cancel Recording

6.5 Recording Storage

Local Storage

- Audio file saved on child’s device

Backend Storage

- Stored in MongoDB (metadata) + file storage

File Naming Convention

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<ChildName>_<StoryName>_<Date>.wav
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Stored Fields

- Child Name
- Story Name
- Date
- File URL / Path
- Duration (optional)

6.6 Admin Module

Admin Login

- Secure authentication system

Dashboard Features

- View all recordings
- Search/filter recordings
- Play recordings
- Download recordings
- Delete recordings

6.7 Story Management (Admin)

- Add new story
- Upload 6 images per story
- Add story title
- (Optional) Add:
- Characters
- Summary
- Instructions

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7. Database Design (MongoDB)

7.1 Collections

Users (Optional)

- name
- createdAt

Stories

- _id
- title
- images (array of 6 image URLs)
- characters
- summary
- instructions
- createdAt

Recordings

- _id
- childName
- storyId
- storyName
- fileUrl
- date
- duration

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8. Technology Stack

Frontend

- React.js
- Tailwind CSS (for child-friendly UI)

Backend

- Node.js + Express

Database

- MongoDB (NoSQL, flexible schema)

Storage

- Cloud storage (AWS S3 / Firebase Storage)

Audio Recording

- MediaRecorder API (browser-based)

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9. UI/UX Design Principles

- Bright and colorful interface
- Large buttons and simple navigation
- Minimal text, more visuals
- Interactive elements for engagement
- Responsive design for tablets and laptops

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10. Security Considerations

- Admin authentication required
- Restricted access to recordings
- Secure storage of audio files
- No public exposure of child data

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11. Scalability Considerations

- Modular backend design
- Cloud storage for large audio files
- Efficient indexing in MongoDB
- Ability to scale to 10,000+ users

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12. Future Enhancements

- Speech-to-text transcription
- AI-based storytelling evaluation
- Gamification elements
- Multi-language support
- Progress tracking for children

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13. Conclusion

This platform provides a structured yet creative environment for children to express themselves through storytelling. By combining intuitive design with robust backend infrastructure, the system supports both user engagement and meaningful data collection for research and education.